

Doubly-Personalised Reading and Listening

Matching Topic and Difficulty to Each Learner

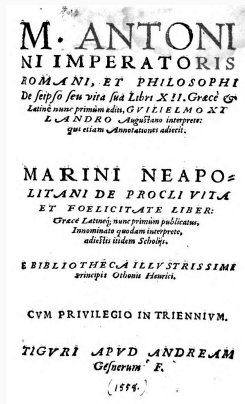
Mircea Lungu · IT University of Copenhagen

IATEFL LTSIG · Green Intelligence · Limassol · 10 October 2026

The best way to learn has been known for almost 2,000 years

“From my great-grandfather, [...] to have had good teachers at home, and to know that on such things a man should spend liberally.”

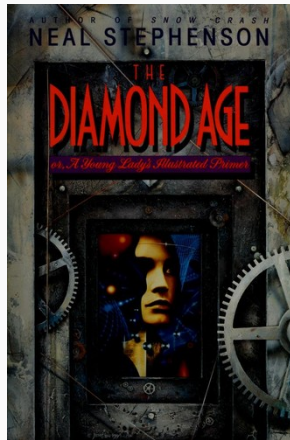
– Marcus Aurelius, *Meditations* 1.4



First printed edition, Zurich 1558

One of my inspirations: *The Diamond Age*

- Nell is a poor girl: no one can afford a tutor for her
- She gets *A Young Lady's Illustrated Primer*, a book that adapts to her



Neal Stephenson, 1995. First-edition cover (Bantam)

Two ways to personalize

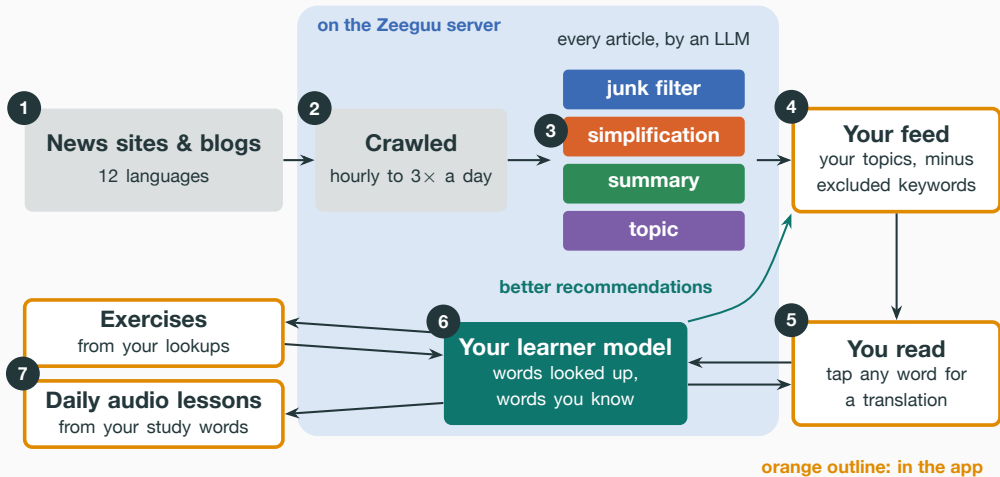
Hypothesis: a learner is more motivated to read (and listen) when the material is

- **about something they care about** → *topic*
- **just above what they can already handle** → *difficulty*

Zeeguu is an app and website that *aims* to do **both**.

Topic: Hidi & Harackiewicz 2000 · Guthrie et al. 1999 · Brozo et al. 2007 · Renandya 2007

How Zeeguu works (simplified)



Install it now

zeeguu.org

or Zeeguu in the App Store / Google Play



Invite code: `1tsig`

Follow along: pick a language you are learning, pick your interests

- ✓ **Topics of interest:** *“Order articles in different subjects like Animals, Politics, Fashion...”*
- ✓ **Keywords to exclude:** *“I would like to avoid articles which have information about accidents with human casualties.”*
- ✓ **Keyword searches:** *“Add a search engine.”*

The screenshot shows a user interface for configuring news preferences. It is divided into two main sections: 'Topics of Interest' and 'Topics to Avoid'.

Topics of Interest: This section contains several orange buttons representing different categories: Business, Culture & Art, Health & Society, Politics, Satire, Sports, Technology & Science, and Travel & Tourism.

Topics to Avoid: This section includes a text input field with the placeholder 'Add a word or phrase to avoid, e.g. robbery' and an orange '+' button to the right. Below the input field are four blue buttons with white text and a close icon (X): trump, bitcoin, blockchain, and Trumps.

At the bottom of the 'Topics to Avoid' section, there is a checkbox that is currently checked, labeled 'Avoid disturbing news (violence, death, disasters)'. Below this checkbox is a small line of text: 'When enabled, articles about violence, war, accidents, and other disturbing topics will be hidden from your recommendations.'

Students' requests, 2017 (CHI 2018). All three are built now.

- Works at CEFR-level granularity
- Titles and summaries are pre-computed
- Full articles are simplified in real time

Your Active Languages:



Language	CEFR Level
English	B1
Danish	A2
German	B2
Spanish	A2
French	A1
Italian	A1
Dutch	A1

Article is above your level

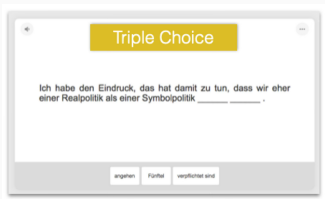


Do you want to adapt it before saving?

Adapt to A2

Demo · Vocabulary Exercises

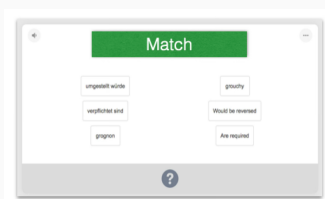
- Eight types in increasing order of difficulty
- Based on the original context (when appropriate)
- A word is considered learned when it's been correct six times in a row



Triple Choice

Ich habe den Eindruck, das hat damit zu tun, dass wir eher einer Realpolitik als einer Symbolpolitik _____.

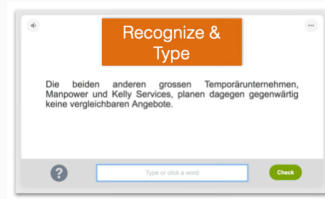
angehen fürht verpflichtet sind



Match

umgestellt würde	grouchy
verpflichtet sind	Would be reversed
gragon	Are required

?



Recognize & Type

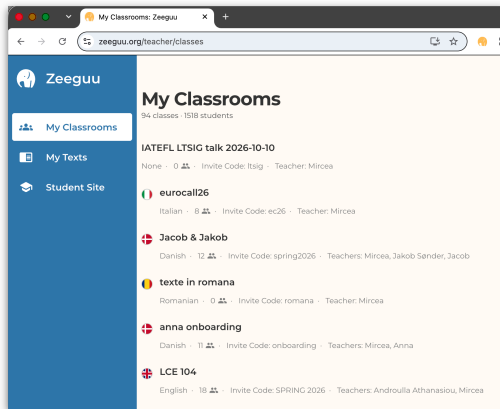
Die beiden anderen grossen Temporärunternehmen, Manpower und Kelly Services, planen dagegen gegenwärtig keine vergleichbaren Angebote.

?

Type or click a word

Check

- Jack and Androulla already explained how to use it this morning
- Social features in the classroom - new feature added by my students



Listening · Daily audio lessons

A dialogue segment

the building block



Vocabulary lesson

2–3 words from your study list



Topic / situation lesson

you name it: “at the taverna”



Under the Hood

- 10 years of uptime
- 10K+ commits
- 100K+ LOC
- 50+ contributors
- **LLMs** (in the last year)

LLMs: The Good, the Bad, and the Awkward

The Good

- simplification to level
- example sentences
- dictionary explanations

The Bad

- difficulty estimation is inconsistent

The Awkward

- simplification to A1 sometimes is awkward
- slow, unreliable (*next slide*)



How we integrate an LLM

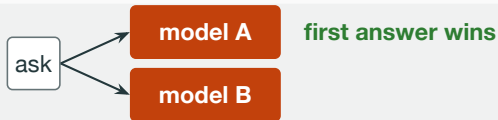
A backup for every model

Fail-Fast Provider Chain



Sometimes: ask two at once

Multiplexed Dispatch



Sometimes: a check gate

LLM-Checking-LLM, Reject and Reprompt



More patterns in the PLoP 2026 paper: <https://llm-patterns.mircealungu.com/paper.pdf>

Zeeguu in the Classroom

“Reading happens at home, in class we talk”

French *and* German teachers at Dutch high schools (a few hundred learners)

Teacher: being able to choose topics is motivating for the students, *“and motivation is critical”*

Students:

- *“Good that you can choose what you read.”*
- *“A very nice website, easy to use and with nice topics.”*
- *“It would be even better if this tool was available in Dutch.”*



Paper (CHI 2018): <https://zeeguu.org/papers/as-we-may-study.pdf>

A textbook personalised by the teacher

Jack Burston and Androulla Athanasiou, CUT Language Centre

- Students see only the texts their teachers upload
- Makes sense for hyper-specialised topics, like their ESP (English for Specific Purposes) university courses

Create any invite code you like (max. 20 characters)

Itsig



Students see only the texts I share

No news feed, search, saved articles or inbox.
Students can still leave the class to see the rest of Zeeguu.



Students see each other

The class appears as a leaderboard for its students. When off, only you see how the students compare.

Open questions for teachers

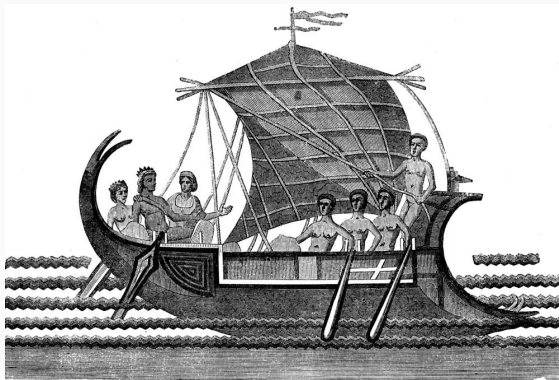
- Are there **other ways of integrating this in the classroom**?
 - How much **choice should students get**, and how much should the teacher keep?
 - What do you **need to see about your students' reading**?
- Would you use the **daily audio lessons**? For which students?
- ***Complexifying*** texts for advanced learners?

Many people have contributed over the years

Konstantina Argyropoulou Martin Avagyan **Androulla Athanasiou** Alin Balutoiu
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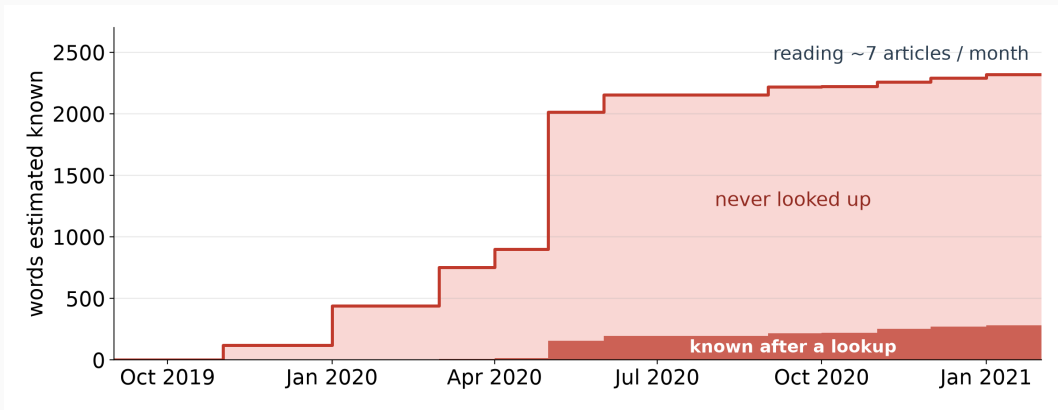
- 10 years of uptime
 - 10K+ commits
 - 100K+ LOC
 - 50+ contributors
 - 10+ theses
-
- zeeguu.org/research
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Theseus's ship leaving Crete, engraving after a Roman mosaic (via scienceabc.com)

- Does it **motivate learners to spend more time**? For whom?
- How do we **measure learning efficiency**?
- How to **personalise further**? E.g. more audio for beginners

Backup · A more personal vocabulary estimate



We infer a learner's vocabulary from how they read: words met again and again but never looked up are probably known.

References

Zeeguu

Lungu, M. F., van der Brand, L., Chirtoaca, D., & Avagyan, M. (2018). As we may study: Towards the web as a personalized language textbook. *CHI '18*. ACM. zeeguu.org/papers/as-we-may-study.pdf

Lungu, M. F. (2026). Estimating receptive vocabulary from incidental reading behavior. *EUROCALL 2026 Short Papers*. zeeguu.org/papers/vocabulary-estimation.pdf

Lungu, M., & Pautasso, C. (2026). Architectural patterns for integrating LLMs into user-facing applications: Lessons from a language-learning platform. *PLoP 2026*. llm-patterns.mircealungu.com

Interest, motivation and reading

Brozo, W. G., Shiel, G., & Topping, K. (2007). Engagement in reading: Lessons learned from three PISA countries. *Journal of Adolescent & Adult Literacy*, 51(4), 304–315.

Guthrie, J. T., Wigfield, A., Metsala, J. L., & Cox, K. E. (1999). Motivational and cognitive predictors of text comprehension and reading amount. *Scientific Studies of Reading*, 3(3), 231–256.

Hidi, S., & Harackiewicz, J. M. (2000). Motivating the academically unmotivated: A critical issue for the 21st century. *Review of Educational Research*, 70(2), 151–179.

Renandya, W. A. (2007). The power of extensive reading. *RELC Journal*, 38(2), 133–149.